

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

One-variable data: Distributions and measures of center and spread

01

10 questions · ~15 min

Q1

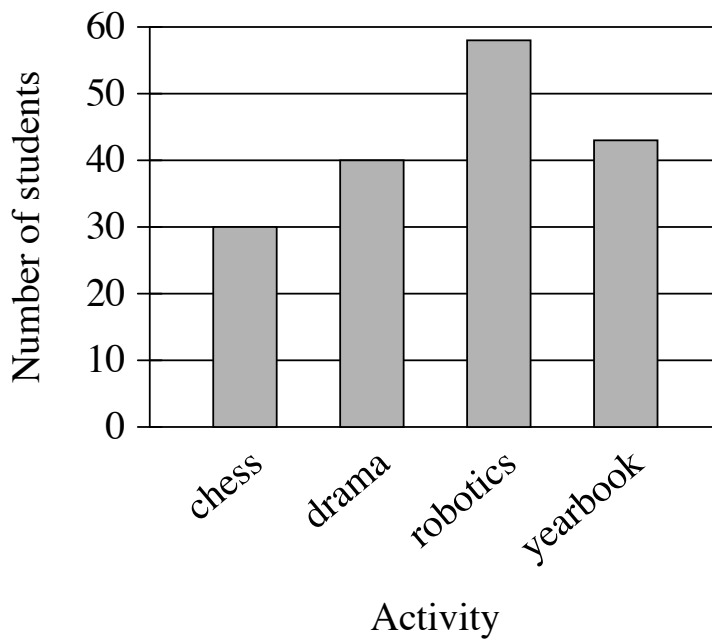
PROBLEM-SOLVING AND DATA ANALYSIS

2, 9, 14, 23, 32

What is the mean of the data shown?

- A. 14
- B. 16
- C. 17
- D. 32

The bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school.



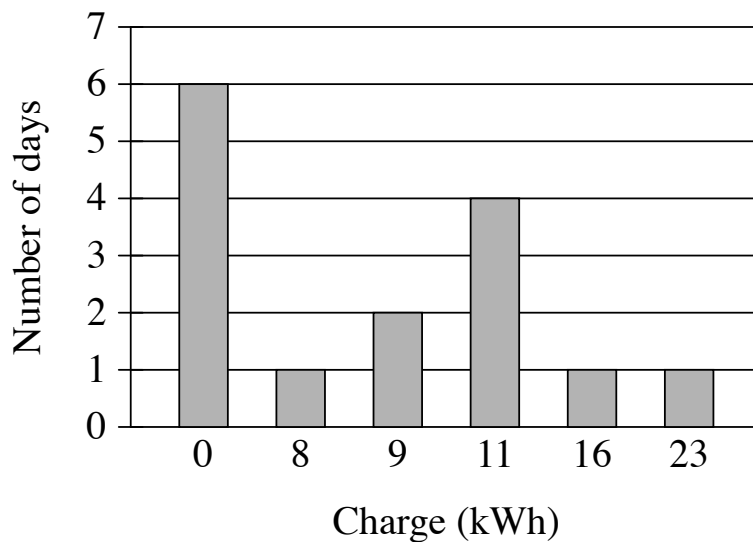
- The Number of students data for the 4 bars are as follows:
 - chess: 30
 - drama: 40
 - robotics: 58
 - yearbook: 43

How many more students are in drama than in chess?

- A. 10
- B. 30
- C. 40

D. 70

The bar graph summarizes the charge, in kilowatt-hours kWh, a battery received each day for 15 days.



- The data for the 6 categories are as follows:
 - 0 kilowatt-hours: 6
 - 8 kilowatt-hours: 1
 - 9 kilowatt-hours: 2
 - 11 kilowatt-hours: 4
 - 16 kilowatt-hours: 1
 - 23 kilowatt-hours: 1

For how many of these 15 days did the battery receive a charge of 0 kWh?

- A. 0
- B. 1
- C. 4

D. 6

Q4

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

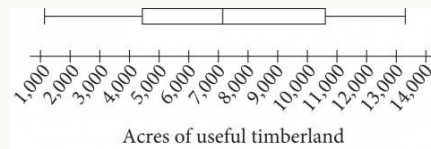
73, 74, 75, 77, 79, 82, 84, 85, 91

What is the median of the data shown?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The number of acres of useful timberland in 13 counties in California is summarized in the box plot above. Which of the following is closest to the median number of acres?

- A. 4,399
- B. 7,067
- C. 8,831
- D. 10,595

PROBLEM-SOLVING AND DATA ANALYSIS

PROBLEM-SOLVING AND DATA ANALYSIS

6, 8, 16, 4, 17, 26, 8, 5, 5, 5

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Five Smallest Countries in 2016

Country	Land area (square kilometers)
Monaco	2.0
Nauru	21
San Marino	61
Tuvalu	26
Vatican City	0.44

The table above shows the land area, in square kilometers, of the five smallest countries of the world in 2016. Based on the table, what is the mean land area of the 5 smallest countries in 2016, to the nearest square kilometer?

- A. 20
- B. 22
- C. 61
- D. 110

4, 4, 4, 4, 8, 8, 8, 13, 13

Which frequency table correctly represents the data listed?

A.

Number	Frequency
4	4
8	3
13	2

B.

Number	Frequency
4	4
3	8
2	13

C.

Number	Frequency
4	16
8	24
13	26

D.

Number	Frequency
16	4
24	8
26	13

Data set X: 5, 9, 9, 13

Data set Y: 5, 9, 9, 13, 27

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

- A. The mean of data set X is greater than the mean of data set Y.
- B. The mean of data set X is less than the mean of data set Y.
- C. The means of data set X and data set Y are equal.
- D. There is not enough information to compare the means.